

SICELUKWANDA ZWANE

🌐 <https://sichelukwanda.com> · @sichelukwanda.zwane@gmail.com

🏠 Cape Town, South Africa

WORK EXPERIENCE

Robotics Researcher and Lecturer, *University of Cape Town – South Africa* **Oct 2026 - Present**

- Lecturing computer science courses.
- Conducting research in robot learning, Vision-Language-Action (VLA) models, sim-to-real transfer, and reinforcement learning.

Robotics Software Engineer - AI/ML, *Kinisi – Bristol, UK* **Apr 2025 - March 2026**

- I implemented and fine-tuned robot foundation models (VLAs/VLMs) for performing real-world tasks on the Kinisi humanoid robots.
- I curated teleoperation datasets for different tasks and applied targeted data augmentation with generative models to improve real-world robustness.
- I developed simulated environments (gymnasium-like interface) for stress-testing the imitation learning pipeline, with support for language-conditioned policies.
- I performed large-scale *simulation-in-the-loop* experiments on the cloud (H100s) to benchmark different VLA approaches, perform hyperparameter sweeps, ablate new vision backbones, etc.

PhD Research Intern - Mobile Robotics, *AIST – Tokyo, Japan* **Jan 2024 - Mar 2024**

- I used Gaussian processes for motion planning in crowded spaces with the lab's mobile robots, namely, a self-driving wheelchair, a bespoke pioneer mobile robot, and a Raspberry Pi Mouse robot swarm.
- I helped evaluate different crowd navigation algorithms in the wild, i.e., at the Miraikan science museum in Tokyo.
- I supervised a masters student to build a world model (recurrent state space model) for modelling pedestrian behaviour.

Postgraduate Teaching Assistant, *University College London, UK* **2020 - 2022**

- Co-supervising student dissertations (term projects) on robotics and reinforcement learning.
- Developing and assessing student coursework in C++ (Robot Sensing, Interaction, and Manipulation).

CTO and Co-founder, *Amathambo AI – London, UK* **2023 - Present**

- I manage the company tech-stack, GCP cloud account, and do hands-on development of Amathambo's AI tool (prototype) for solving medical staff scheduling problems.
- I set up and own the CI/CD pipeline with Github actions for Amathambo's flagship product, Manje and company website.
- I train autoregressive models for predicting hospital patient load. These are integrated into the web app by our backend developer.

Data Scientist, *Explore AI, South Africa* **2019 - 2020**

- I led a team that developed a text Tweet sentiment analysis tool and accompanying dashboard for a big South African bank.
- I developed a 6-month machine learning training course for explore AI academy. The course included capstone projects for regression, classification, and unsupervised learning.

Research Intern - Field Robotics, *CSIR – Pretoria, South Africa* **2017 - 2019**

- I developed and maintained a ROS-based software stack for the perception and control of a mobile manipulator robot.
- I developed a simulated environment for the physical mobile manipulator robot in Gazebo to bootstrap reinforcement learning experiments.

EDUCATION

PhD Robotics and Artificial Intelligence, *University College London (defended)* **2020 - Present**

Highlights: I developed a BayesOpt soft robot control learning algorithm for throwing and hammering that outperformed a modern reinforcement learning agent (REDQ).

MSc Computer Science, *University of the Witwatersrand* **2017 - 2019**

Highlights: I showed that we can learn faster and safer in reinforcement learning by leveraging a mixture of expert actions model for exploration.

BSc (Hons) Computer Science (With Distinction), *University of the Witwatersrand* **2013 - 2016**

Highlights: I built a Mobile Telepresence Robot that used the ROS Indigo navigation stack for SLAM, ROS HTTP Bridge for the UI, and Mumble for VoIP.

SKILLS SUMMARY

Machine Learning	Imitation Learning, Reinforcement Learning (model-based, safe exploration), Probabilistic ML (modeling robot dynamics, motion planning, Bayesian optimisation)
Generative AI:	Flux, SDXL, Diffusion-based Data Augmentation, Depth Estimation
Simulation:	Mujoco (+ dm-control), ManiSkill, PyBullet, Genesis, Gazebo
Computer Vision:	Vision Transformers, CLIP, 3D Vision (Depth Estimation), Gaussian Splatting, Semantic Correspondences, Scene & Object Segmentation (SAM2)
Frameworks:	PyTorch, JAX, HuggingFace, Weights & Biases, Hydra, TensorFlow
Training & Ops:	Distributed Training (H100 Clusters), SkyPilot, GCP, AWS, Docker, Singularity, GitHub Actions (CI/CD)
Languages:	Python, C/C++

SELECT PUBLICATIONS

Semantic Cross-Pose Correspondence from a Single Example, ICRA 2025

Hadjivelichkov D, **Zwane S**, Deisenroth MP, Agapito L, Kanoulas D

The paper presents a method for predicting a semantically corresponding pose of a pair of unseen objects, given one or a few example poses showing the interaction of a similar object-pair.

Learning Dynamic Tasks on a Large-scale Soft Robot in a Handful of Trials, IROS 2024

Zwane S, Cheney D, Johnson CC, Luo Y, Bekiroğlu Y, Killpack MD, Deisenroth MP

Developed a data-efficient learning-based controller for a large-scale soft robot using Bayesian optimization. Evaluated on hammering and throwing tasks in simulation (Mujoco) as well as on a physical soft robot.

Safe Trajectory Sampling in Model-based Reinforcement Learning, CASE 2023

Zwane S, Hadjivelichkov D, Luo Y, Bekiroğlu Y, Kanoulas D, Deisenroth MP

Learned a constraint-aware Gaussian process dynamics model, trained a safe policy on trajectories sampled from it, and deployed the resulting safe policy on a physical robot arm. Pybullet and real-robot evaluation.

AWARDS AND ACHIEVEMENTS

2025 – Global Talent Visa

UK Visas and Immigration

A visa that allows endorsed leaders or emerging leaders in academia and research, arts and culture, or digital technology to live and work in the UK (Valid till 2030).

2023 – Äänit Prize

Nelson Mandela Rhodes Foundation

Amathambo AI secured \$38,000 in funding to scale a proof-of-concept that matches hospital staffing levels to demand using machine learning.

2023 – Research Poster Award

Deep Learning Indaba

Awarded a book for a research poster at the Deep Learning Indaba 2023 in Ghana.

2020 – DeepMind Full-Scholarship for PhD in Foundational AI

University College London

Awarded £125,000 full funding towards the pursuit of PhD in Robotics and AI.

2019 – Machine Learning Summer School (MLSS) Travel Scholarship

Element AI

Awarded approx. \$3000 to attend MLSS London, covering flights and accommodation.

2018 – Google Cloud Credit Award

Deep Learning Indaba

Awarded \$1000 in Google Cloud Platform credits for a research presentation at the Deep Learning Indaba 2018.

2017 – NeurIPS 2017 Praekelt Travel Award

Deep Learning Indaba

Awarded the travel prize for best poster at the Deep Learning Indaba 2017 to attend NeurIPS 2017 Conference in Long Beach, California, USA.

2017 – ROSCon 2017 Diversity Scholarship

ROSCon 2017

Received a travel scholarship to attend ROSCon (Robotics Operating System Conference) in Vancouver, Canada.

2017 – NRF MSc Scholarship (declined)

National Research Foundation

Awarded approx \$2500 towards the pursuit of a masters degree.

2016 – SKA Masters Bursary (declined)

Square Kilometer Array

One of 10 individuals across 3 countries in Southern Africa to receive the SKA bursary for Masters studies. (approx. £20,000 over 3 years)

2016 – Most Innovative Honours Research Project

University of the Witwatersrand

Awarded a cash prize of approx. \$50 for building a semi-autonomous mobile telepresence robot with face-tracking and autonomous navigation.

- **Founding Member**, Akili African Robotics Community (AARC): Co-founded a pan-African initiative uniting researchers, roboticists, and engineers to expand access to robotics education, hardware, and research across the continent.
- **Workshop Organiser**, Robotics & Automation in Africa: Co-organized the flagship robotics workshops at the annual Deep Learning Indaba across Africa, including Ghana (2023), Senegal (2024), and Nigeria (2026).
- **Tutorial Lead & Open-Source Developer**: Designed and led the hands-on VLA Robot Learning Tutorial (Action Chunking with Transformers) at DLI 2026; core contributor to the open-source *zandla* robot learning framework for the SO-101 robot arm.
- **Organiser**, IndabaX South Africa: Co-organized national machine learning indaba workshops to support African ML researchers and build local capacity.
- **Academic Reviewer**: Reviewer for robotics and machine learning venues and workshops (ICRA, IROS, NeurIPS Black in AI workshop).